Research on Generative Artificial Intelligence in Senior High Schools' English Reading Teaching

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Abstract. The advent of generative artificial intelligence (GAI) and the prominence of core literacy in senior high school have jointly promoted innovative transformations in teaching. This paper employs literature review methodology to first examine current problems in senior high school English reading class, subsequently proposing GAI' s integrative advantages. Then further identifies implementation challenges and corresponding countermeasures. It is found that traditional English reading class has problems of single reading resources, lacking thinking training and one-sided reading evaluation system, while GAI can enrich reading materials, promote critical thinking, and achieve diversified evaluation. However, there are problems of lacking emotional interaction, teachers' technical dependence and difficulties in changing teaching concepts. Therefore, this paper proposes a teacher-led, GAI-assisted emotional interaction mechanism, teacher-student-AI triadic collaboration, and progressive conceptual iteration strategies. The paper shows that the deep integration of GAI and English reading teaching needs to balance technological empowerment and traditional educational values.

1 Introduction

1.1 Background of the study

The English Curriculum Criteria for Regular High School (2017 Edition, 2020 Revision) integrates language, culture, and thinking organically within the core competencies of English as a discipline, establishing concrete and feasible academic goals to achieve moral education and disciplinary education[1]. As a pivotal component of senior high school English teaching, reading teaching not only enables teachers to convey information, enhance students' language proficiency and cultural literacy but also serves as a platform for cultivating students' thinking abilities. However, traditional English teaching faces challenges such as slow improvement in students' critical thinking, superficial cultural understanding, and a lack of diversity in reading materials.

Since OpenAI launched ChatGPT in October 2022, generative artificial intelligence (GAI) has entered mainstream awareness. By 2024, the convening of the World Digital Education

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Conference has brought unprecedented attention to the application of GAI in education. Due to its powerful language processing and content generation capabilities,
GAI can help innovate the way English reading is taught and improve teaching efficiency and student learning outcomes.

1.2 Literature review

Since the advent of Chatgpt, GAI has gained widespread attention across academic disciplines. Through their research, Bordas et al. demonstrated that GAI can generate entirely new datasets—such as text, images, audio, and video—based on existing data[2]. Jauhiainen and Guerra stated that ChatGPT-3.5 and 4 can effectively customize educational content to provide students with personalized, interactive learning materials and environments[3]. For example, in a senior high school geography classroom, the Kimi intelligent assistant has been shown to optimize instructional design by automating low-level, repetitive evaluation tasks for teachers and tailoring personalized learning content for students[4]. In English translation teaching, GAI can break spatial and temporal constraints to achieve personalized and differentiated teaching, thereby improving the efficiency of translation teaching[5]. Hao et al. proposed a framework for designing personalized teaching pathways through AI technology, which can be applied to senior high school science education to achieve precision teaching[6].

Scholars have also focused on integrating GAI into senior high school English teaching. Wu points out that English teachers can make reasonable use of GAI in teaching design, lesson preparation, teaching assistance, assessment processes and student engagement, so as to improve the quality of teaching[7]. Similarly, Xiang and Li argue that AI empowers teachers to provide richer curricular resources, enabling “large-scale, personalized” precision teaching[8]. Meanwhile, students can develop self-directed learning skills through AI-driven English learning applications or platforms. Furthermore, Chen proposes that GAI-enhanced reading practices, mediated through human-AI dialogue, allow readers to efficiently acquire, process, and comprehend information while fulfilling demands for intelligent and personalized reading experiences[9].

In summary, scholars globally acknowledge the significant role of GAI in education, providing a robust theoretical foundation for its integration into English teaching. However, Rahman and Watanobe cautioned that despite ChatGPT’s utility in educational research, it remains constrained by limitations such as a lack of common sense, potential biases, difficulties in complex reasoning, and an inability to process visual information, arguing against uncritical reliance on the technology[10]. Additionally, Zhang and Wang highlighted challenges in digital reading, including data-driven limitations, superficial comprehension, and passive information consumption. To address these issues, they emphasized the need for teachers to focus on cultivating students’ ability to actively select resources, enhance human-AI interactivity to avoid homogenization, and develop critical textual analysis skills in classroom settings[11].

1.3 Research topic

Existing studies still exhibit critical gaps. While current studies point out GAI’s potential ability in the educational field, such as assisting in teaching design and realizing personalized teaching, etc. They also put forward its limitations and coping strategies in the teaching process. However, there is a lack of in-depth and systematic research on how to use GAI in a concrete and rational way to address the problems in traditional senior high school English reading classrooms.

Therefore, this paper aims to explore the effective path of integrating GAI tools into senior high school English reading teaching class. The findings will offer practical cases and theoretical support for the deep fusion of educational technology within subject-specific teaching, thereby advancing the modernization of education.

2 Problems in the traditional senior high school English reading classroom

2.1 Homogeneity of reading materials

Most teachers believe that textbooks are the syllabus of English language teaching[12], with classroom content mainly centered on prescribed materials. However, textbooks materials mostly focus on academic genres such as expository, argumentative, and narrative essays, and lack authentic language materials such as original novels, overseas social media, and TED talks.

Furthermore, the updating rate of textbooks is much slower than the development of the times, resulting in outdated and contextually irrelevant content[13]. For example, the People's Education Press textbooks, which are the most widely used in China, were compiled in 1984, 1996, and 2003[14]. The latest version textbook was not until 2017 that began to be written, and it was gradually put into use in 2019. The sluggish revision pace fails to incorporate real-time hotspots such as artificial intelligence and digital economy in a timely manner, resulting in a serious disconnect between classroom knowledge and contemporary realities. Consequently, students are unable to access diverse cultures and global dynamics through reading, limiting their preparedness for an interconnected world.

2.2 Neglect thinking training in the reading process

Reading teaching serves as a critical pathway to promote students' thinking progress[15]. However, fixed teaching hours and limited teacher energy often lead teachers to dominate classroom discourse to complete curricular tasks on schedule. Consequently, students passively absorb the general meaning of the text, vocabulary, grammar and other knowledge, with minimal opportunities to question, discuss and share their own opinions on the text. This not only fails to stimulate students' critical questioning or personalized interpretation skills, but also makes reading lessons monotonous and reduces students' motivation to read deeply and think independently[16]. For example, in cross-cultural discourse teaching, teachers only introduce the cultural background with the help of images and audio[17], failing to guide students to conduct classroom discussions, comparative analysis of cultures of different countries and other thinking inquiry activities[18]. Such practices not only lead to students' shallow understanding of different culture, but also lead to a serious lack of critical training.

2.3 One-sided evaluation system of reading

Traditional assessment methods often overemphasize accuracy rates, relying solely on standardized test scores to evaluate students' reading comprehension abilities, without accounting for individual differences in language ability[19]. For example, most of the exams are based on objective questions such as multiple-choice questions, which narrowly focus on students' comprehension of factual grasp in the text and mastery of vocabulary and grammar. This approach neglects the effective assessment of students' stage-by-stage progress, such as reading fluency, logical reasoning, cross-cultural comparisons, and argumentation skills.

This makes teachers very difficult to make accurate and effective adjustments without a comprehensive understanding of the students' learning process.

3 Advantages and Challenges of GAI into Senior High School English Reading Instruction

3.1 Advantages

3.1.1 Enriching reading instructional resources

GAI overcomes the physical limitations of printed textbooks, enabling students to access a wider range of text genres. For example, through an algorithm-driven text generation mechanism, GAI can quickly present the literary metaphors in the original novels, the localized expressions in social media, and the academic colloquial features in TED Talks. Thereby, this enables students to become familiar with informal, conversational, and contemporary language used in daily digital communication and enhances their capability to use of language in real life.

In addition, with its neural network-based trend tracking systems, GAI further eliminates the time difference barrier between the textbook content and the real world. By dynamically capturing emerging concepts and sociocultural issues from global knowledge networks, it converts these into scalable and extensible reading modules, ensuring students remain tightly aligned with current realities.

GAI can further realize personalized reading. Through analyzing of students' reading habits, thematic preferences, and language proficiency levels, it curates tailored content to individual interests. For students interested in science and technology, GAI can recommend reading materials related to the cutting-edge scientific research, technological inventions, and space exploration. For those who love culture, it can provide materials about different national cultures, traditional festivals and art forms. This personalized approach greatly increases the student's engagement with the reading material, making the learning process more engaging and productive.

3.1.2 Fostering critical and dialectical thinking

In senior high school English reading classes, teachers can design layered interactive tasks through GAI to exercise their thinking skills. Firstly, in the stage of in-depth text analysis, GAI generates a chain of questions targeting foundational, analytical, and evaluative dimensions of key passages. At the same time, GAI is requested to provide examples of analysis from different perspectives, such as historical background, literary theory, reader response criticism, etc., and to make comparative analyses with the students' own perspectives to explore difference in understanding.

Subsequently, in the critical dialogue stage, GAI is configured in the opposing position. Through its powerful question-and-answer function[20], students are encouraged to debate with it using textual evidence. In this process, GAI identifies in real time logical gaps in the student's arguments (e.g., "Your argument lacks specific examples from paragraph 3") and provides suggestions for corrections. This model not only preserves students' subjective exploration space, but also transforms fragmented reading behaviours into systematic thinking trajectories through intelligent cognitive interventions.

3.1.3 Diversifying evaluation system

GAI provides innovative technical support for the diversification of the English reading evaluation system[21]. It can transcend the limitations of traditional standardized tests and provide students with multi-dimensional and dynamic assessment models. Firstly, GAI can generate questions and quizzes according to different difficulty levels of the same topic[10]. Teachers can leverage this capability to assign homework and control the difficulty level, and increase the complexity of the text and difficulty of the tasks in steps based on students' proficiency levels. This ensures alignment with the zone of proximal development while mitigating cognitive overload.

At the same time, GAI can also record students' reading behaviour data in real time through natural language processing and big data analytics to provide comprehensive evaluation. For example, when students are reading, GAI can not only judge the accuracy of the answers, but also record the students' reading speed, pause time, and frequently revisited content, etc. These data can visually reflect the difficulties students encountered during the reading process and the attention they paid to different content. When students answer open-ended questions or write about their reading conclusions, GAI can quickly identify logical holes, weak argumentation, cultural misinterpretation and then provide targeted advice. This facilitates a multidimensional assessment spanning vocabulary-grammar mastery to critical thinking proficiency. By analyzing such process data, teachers can adjust their teaching strategies in time to provide more targeted guidance to students. Crucially, evaluation can no longer be limited to stage exams, but throughout the entire learning process.

3.2 Challenges

While GAI can partially address the shortcomings in the traditional senior high school English classroom, its potential risks still cannot be ignored.

3.2.1 Deficiency in emotional education

Although GAI can be trained on large textual datasets to generate human-like texts, the essence of their emotional expression is an algorithm-driven combination of symbols. This limitation becomes evident in human-AI interaction, particularly when addressing students' individualized emotional needs[22]. Since the traditional models only centered on single modalities like video or audio, which restricts their capacity to capture the complexity of human emotions conveyed through multiple channels[23], they often fail to sense students' emotional fluctuations and respond them in a timely and adaptive manner. As Sharma et al. noted, "Human sentiment in natural language is generally an intricate combination of emotions, which can sometimes be indeterminate, neutral, or ambiguous[24]." Therefore, students may be less engaged in learning due to the lack of non-verbal feedback such as teachers' encouraging gestures, tone modulation, or empathetic cues. For example, after a student answers a question, a teacher might show encouragement to that student through body language and expression, whereas GAI's feedback is confined to textual responses based on pre-trained data. Fundamentally, GAI lacks the capacity to establish neural correlations between linguistic symbols and real-world sensory experiences or emotional memories. This disconnect means that its comforting discourse operates only at the level of semantic logic, missing the dynamic responsiveness to paralinguistic elements like dialogue context and interpersonal relationships. Such oversights can lead to overly simplistic, misunderstood, or indifferent interactions, ultimately leaving students unable to derive emotional satisfaction or motivation from the learning process.

Furthermore, modern teacher evaluation frameworks emphasize that high-quality educators must not only improve academic performance but also support students' socio-emotional development[25]. In the process of teaching reading, the emotions expressed by the author in a discourse are context-sensitive. However, as mentioned above, AI models trained on large-scale corpora tend to reduce complex cultural emotions to probabilistic statistics. Compared with teachers' explanations based on their own deep understanding of the text and unique insights, GAI interpretations lack that emotionally resonant power that can touch people's hearts. Students thus find it challenging to connect empathetically with the affective dimensions of texts, hindering their ability to internalize cultural sentiments or cultivate emotional intelligence.

3.2.2 Teacher overreliance on technology

While GAI can provide teachers with vast teaching resources, including lesson templates, detailed lesson plan designs, reading materials, and diverse exercise formats, etc., it makes some teachers overly rely on intelligent tools for lesson preparation. Thereby, teachers' motivation to deeply engage with or critically reflect on curricular content will be diminished. During class, teachers can easily use interactive software to ask questions, take quizzes and other activities. However, this reliance may lead to a certain degree of decline in teachers' basic teaching skills and weakening of their language skills. During the evaluation stage, overreliance on GAI for assignment creation and grading further exacerbates this issue. Teachers rely too much on GAI to set and correct assignments, and lack of personalized comments. Consequently, teachers may fail to holistically assess learning progress or provide targeted guidance and encouragement, undermining their ability to nurture students' academic and emotional growth.

3.2.3 Difficulty in changing teaching concepts and methods

While some teachers are able to use GAI for teaching, there are still some teachers, particularly veteran teachers, who struggle to abandon entrenched traditional teaching methods. GAI-driven classrooms prioritize student-centered learning, meaning construction, and multidimensional evaluation, which is contrary to the traditional teacher-led classroom. They are concerned that the emergence of GAI will lead to students' cursory mastery of topics, fragmented learning, and thus lack of an overarching knowledge framework [26].

A research survey in China found that almost 1/3 of the teachers interviewed were found to have minimal understanding of ChatGPT at all, and all of them were primary and secondary school teachers[27]. Due to lack of technical expertise and practical experience, they do not know how to use GAI to aid reading comprehension, nor do they know how to effectively integrate GAI-generated content into classroom activities.

Therefore, even though some teachers use GAI to expand materials and personalize interactions, the classroom still tends to be in the traditional mode. For example, they may use texts generated by GAI directly, but lack specific strategies to guide students in analyzing, critiquing or interacting with the text. When students encounter vocabulary and difficult sentences in reading, teachers do not know how to guide students to use GAI to obtain real-time detailed paraphrases and grammatical analyses. They also do not grasp how to use GAI to present sentence structure and text logic in a more intuitive form such as diagrams and mind maps. Instead, they rely on limited explanatory methods like looking up the words in the dictionaries and using blackboard writing to analyze sentence composition, resulting in reduced comprehension and ineffective classroom interactions.

4 Strategies for Integrating GAI into Senior High School English Reading Teaching

4.1 Teacher-led, AI-assisted emotional interaction mechanism

Teachers should actively participate in the process of human-AI dialogue and pay close attention to students' emotional states. When students are communicating with GAI, teachers need to step down from the podium and observe students' learning progress and changes in their expressions from a close distance. When teachers detect signs of confusion or frustration such as furrowed brows and prolonged hesitation, they should intervene in a timely manner to ask students what their current confusing points are and give them academic guidance and spiritual encouragement, such as affirming eye contact, empathetic tone modulation, and encouraging body language. After students complete the reading task with the help of GAI, teachers can reinforce motivation with immediate smiles, nods or verbal praise to enhance students' satisfaction and sense of achievement.

After all students have completed the human-AI interaction, teachers should reinterpret the emotional dimensions of texts based on their deep understanding and personal perception of the text. By sharing their own emotional experiences, students are guided to internalize the emotions expressed by the author in a specific context, helping them to resonate with the emotions of the text. For example, when analyzing an English reading article about friendship, teachers can combine their own experiences to share the sweet and sour of friendship, so that students can more viscerally understand the emotions expressed in the text, rather than relying on the simple probabilistic statistical interpretation given by GAI.

4.2 Teacher-Student-AI tripartite collaboration

In the reading class, teachers should act as the guide of cognitive construction, students as the main body of active inquiry, and GAI as a dynamic enabling technological mediator, jointly building a collaborative teaching with adaptability, innovation and sustainability. In the stage of lesson preparation, teachers should first conduct a comprehensive and in-depth study of the English reading textbook to clarify the teaching objectives, key points and difficult knowledge. Then, teachers need to design the whole teaching process according to their own teaching experience and teaching style. In this process, teachers should give full play to their own initiative and independently complete the preliminary framework of the teaching design by virtue of their own teaching experience, class characteristics, and knowledge level. After completing the preliminary framework of independent lesson preparation, teachers can use GAI to generate multiple thematic texts as reading materials. These materials need to be filtered by Flesch-Kincaid Readability Index to quantify linguistic complexity and ensure alignment with students' proficiency. For example, the average readability level of People's Education Press high school English texts ranges between 5.7 and 8.4[28]. GAI-generated texts within this range are prioritized for relevance and accessibility. However, the consensus of the researchers is that the data is not enough to fully describe complexity. For example, they believe that longer sentences are grammatically more complex than shorter ones, which may not always be true[29]. Therefore, a combination of subject matter expert review is needed to finally select the suitable article.

During the teaching process, students participate in teacher-guided group readings of GAI-generated texts, and analyze GAI-generated text by asking questions, annotating or discussing it in groups. Key evaluation criteria encompass informational completeness, logical coherence, and factual accuracy. A study with 90 high school students in Indonesia concluded that the guided inquiry-based learning model contributes to improving students'

critical thinking skills[30]. When students encounter vocabulary and difficult sentences that hinder their reading, they can ask GAI to explain complex sentence structures, add cultural background, or put forward opposing viewpoints. Concurrently, teachers need to observe the student-AI dialogue, and intervene in a timely manner to point out the logical loopholes or linguistic blind spots. For example, when the system detects that students have asked questions about the same grammatical difficulty for three consecutive times, GAI should automatically push the micro-teaching video or when GAI gives the cultural background, the teacher can insert guiding questions such as "Is there any bias in the cultural background given by the AI? Please compare the illustration on page X of the textbook" for comparative thinking. Through this framework, students synthesize autonomous reasoning, GAI-generated conceptual scaffolds, and teacher-mediated critical analysis to achieve deep learning—progressing from information extraction to perspective reconstruction. In this model, GAI acts as a content generator and instant feedback source, students become active explorers and reflective practitioners, and teachers focus on higher-order thinking guidance and collaborative process regulation, forming a closed loop of "resource-practice-sublimation", taking into account the dual cultivation of linguistic competence and digital literacy.

4.3 Progressive conceptual iteration

Teachers need to recognize that the future of education requires a harmonious integration of knowledge, skills, attitudes and values. GAI emerges as teachers' strategic ally, enabling teachers to enhance learning experiences while equipping students for a future where AI will constitute an essential component of problem-solving and innovation[31]. Therefore, teachers need to be optimistic about it, embrace it and utilize it.

Teachers can start with small-scale practical teaching experiments, from the simplest activities such as using GAI for lesson preparation assistance or designing basic interactive classroom activities. Such limited technological integration can both reduce cognitive load and develop teachers' ability to critically integrate technology in practice. After the lesson, teachers must systematically document instructional strengths and weaknesses, solicit student feedback regarding learning difficulties and GAI-assisted reading experiences. Based on these feedbacks, the teaching content and activity design should be optimised and adjusted. At the same time, teachers should actively observe excellent AI-integrated English reading courses, learn the skills of others in the articulation of teaching links, student guidance and the use of AI tools, and then skillfully apply the experience accumulated in the observation to their own subsequent teaching activities, so as to gradually realize the change of teaching concepts and improve the quality of English reading teaching.

5 Conclusion

Taking senior high school English reading teaching as an entry point, this paper systematically explores the application path of GAI in teaching through literature review and strategic analysis method. Addressing prevalent issues in traditional classrooms, including homogeneous reading materials, insufficient thinking training, and one-side assessment frameworks. This paper accordingly explores GAI's potential to diversify resources, facilitate critical and dialectical learning, and enable multidimensional evaluation, while also identifying challenges including emotional disengagement risks, technological overreliance and teaching concepts and methods changing difficulty, with corresponding mitigation strategies.

The paper found that GAI can dynamically generate interdisciplinary and multimodal real-time corpus, make up for the lag of teaching materials, and promote the development of

students' critical thinking and intercultural competence through intelligent Q&A and interactive scene design. In addition, the process evaluation supported by GAI can comprehensively record the data of students' reading behaviour and provide a basis for accurate teaching. However, GAI also has certain limitations. In the senior high school English classroom, it suffers from problems such as mechanisation of emotional feedback, teachers' technological dependence and the solidification of traditional teaching concepts, which may weaken the effectiveness of its application. In this regard, the study proposes three strategies: first, to build a teacher-led, AI-assisted emotional interaction mechanism to make up for the emotional deficiencies of GAI through humanistic care; second, to establish a ternary collaboration model of "teacher-student-AI" with a clear division of roles among all parties, realizing a closed-loop of resources, practice and sublimation; and third, to advocate progressive technology iteration, and promote the transformation of teaching philosophy through small-scale practice and continuous reflection.

However, this study has some limitations. Although it provides a practical path for the innovation of senior high school English reading teaching, it still stays at the theoretical level and does not verify its feasibility in practice. In the future, it is hoped that it can be practiced in specific reading teaching classrooms to improve and optimize the problems exposed.

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